Lesson notes · for you

# Finding groups

Data science · Level 4 · Lesson 2

**Time**
30 min

**Materials**
A notebook and a pencil; A phone or tablet with Darsi, for the lab

## Learning goals

- Tell clustering from classification.
- Run k-means steps.
- Use groups with care.

## Activity: Human k-means

1. Warm-up (5 min): Look at a scatter of dots with no colours and circle any groups you see.
2. Move (10 min): Two classmates stand as centres and everyone walks to the nearer one.
3. Repeat (10 min): Each centre moves to the middle of its group, and everyone checks again.
4. Wrap-up (5 min): Say your finding in one sentence with a number in it, then fill in the exit ticket.

## Discussion

- Did anyone change group?
- Does clustering need labels?
- Why choose neutral names?

## Check yourself

- Centres at 3 and 10. Which centre does a pupil with 6 loans join?

**Try it this week**
Plot the sleep and screen time of a few friends, without names, and look for groups. Give each group a kind, neutral name.

Worksheet

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Name _____ Class _____ Date _____

**1** **Does clustering need examples that already have labels?**

◯ Yes, always ◯ No, it finds groups on its own ◯ Only for k = 1

**2** **Order one run of k-means.**

☐ Repeat until assignments stop changing

☐ Move centroids to their means

☐ Assign points to the nearest centroid

☐ Choose k and initial centroids

**3** **Run one k-means step in 1-D with centroids 2 and 5, and print the new centroids.**

```
x = [1, 2, 3, 8, 9, 10]
c = [2, 5]
g = [[], []]
print(c)
```

Write your program:

**4** **k-means always finds the globally best clustering.**

◯ True ◯ False

**5** **Within-cluster error is 100, 40, 32, 29 for k = 1, 2, 3, 4. Which k does the elbow suggest?**

◯ 2 ◯ 1 ◯ 4

✎ **Apps put people into groups to show them adverts. Which group do you think an app has put you in?**